Where do introduced plants add butterfly hosts?

Earlier work documented alien host use in California and naturalized plant richness on islands. Here, each butterfly keeps its known hosts. A region gains potential resources only when an introduced plant supplies its first known host for that butterfly.

A Global resource geography

239 butterflies × botanical regions
126 insular and 237 mainland regions

Comparison: native plant ranges versus native plus introduced ranges.

Gain in butterfly–region combinations

Islands 4,829 → 10,570

+118.9%

Mainland 21,215 → 29,901

+40.9%

B Vegetation near ports

1,596 ports with plant inventories (5 km)
687 candidate introduced host species

Regional comparison: port counts and introduced flora in 98 island regions.

Ports with a candidate host record

1,150 / 1,596 (72.1%)

Port-rich regions had more host taxa. Accounting for all introduced plants removed the positive host association.

C New island butterfly records

27 islands × 132 butterflies, 2011–2024
42,228 at-risk species–island–years

Exposure: a compatible candidate host recorded near a port in 2001–2010.

Annual first-record rate on the island

Earlier host 66 / 5,675

1.16%

Unconfirmed 200 / 36,553

0.55%

Adjusted annual difference **+0.35 percentage points (95% interval +0.03 to +0.68)**

Adjustment includes year, distance to an earlier same-species island, island and butterfly species. The three samples share host identities. Ship transport, arrival and breeding were unobserved.